Biochemistry Cumulative Exam
April 27, 2012

Complex I is an integral membrane protein in the mitochondrial respiratory chain which transfers two electrons from one molecule of NADH and 2 protons from the inside of the membrane to one molecule of oxidized ubiquinone (Q) to form reduced ubiquinol (QH_2) and pumps 4 protons from the inside to the outside of the mitochondrial membrane.

$$NADH_{in} + Q + 5\ H^+_{in} \rightarrow NAD^+_{in} + QH_2 + 4\ H^+_{out}$$

NADH has a redox potential of -0.32 V, and Ubiquinone has a redox potential of $E_o' = +0.10$ V. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.1$ V, and the pH inside and outside the mitochondrial membrane was $pH_{in} = 8.0$, and $pH_{out} = 6.0$.

1. (35) Give a diagram showing how complex I couples the energy of electron transfer to pumping protons across the mitochondrial membrane, and briefly discuss the mechanism recently proposed by Sazanov et al.

2. (25) Calculate the free energy change ΔG^o for electron transfer from 1 mole of NADH to one mole of Q. (not including the energy of proton pumping).

3. (30) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane. Calculate ΔG for the proton pumping component of the complex I reaction under the above conditions. You can assume that the energy for transporting two protons from the inside of the membrane to Q is zero.

4. (10) Calculate how efficiently complex I couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping under the above conditions.

Possibly useful formulas:

$\Delta G^o = - RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$; $\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

Membrane Transport: $\Delta G(X_{in} \rightarrow X_{out}) = RT\ln([X_{out}]/[X_{in}]) + ZF(V_{out} - V_{in})$.

Z = charge on X

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol